Certificate

OF APPRECIATION

Computer Vision-Based Cloud Radar System for Solar Power Generation

This is to certify that the project team cited in this certificate of appreciation representing the faculty and students of the Artificial Intelligence & Data Science (AI&DS) Department at Kumaraguru School of Innovation (KSI), Kumaraguru Institutions, have successfully developed a computer vision-based Cloud Radar System (CRS) for enhanced solar power generation in collaboration with the Kumaraguru Centre for Industrial Research & Innovation and M/s Ezon Solar Tracker (P) Limited, Coimbatore.

The CRS identifies the brightest spot in the sky during cloudy conditions and determines the optimal east-west and north-south angles for the solar panel tracking system. These calculated angles guide the tracking system, ensuring precise positioning of solar panels for maximum sunlight capture.

The developed system integrates a high-resolution camera, advanced image processing software, and an intuitive user interface for real-time monitoring and historical data analysis. Key features include:

- *Extensive sky coverage for comprehensive cloud pattern analysis.*
- *Precise brightness detection to identify optimal sunlight exposure points.*
- *Real-time processing for immediate adjustments to the solar panel orientation.*

By dynamically optimizing the angle and orientation of solar panels based on real-time weather conditions and astronomical events, the system ensures maximum solar energy capture throughout the day and across seasons. This innovation significantly enhances energy generation efficiency.

The computed angles obtained using the proposed computer vision solution were validated against solar positioning data received from M/s Ezon Solar Tracker (P) Limited, ensuring accuracy and reliability.

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